Data Mining

Evaluation of Clustering

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Major tasks of clustering evaluation are

- *Assessing clustering tendency*
  - Assess whether a non-random structure exists in the data.
  - Clustering is meaningful only when there is a nonrandom structure in the data.
  - Can use different statistical method to check randomness such as **Hopkins Statistic.**

Evaluation of Clustering

- *Determining the number of clusters in a data set.*
  - Often depends on the distribution's shape and scale in the data set, as well as the clustering resolution required by the user
  - Some simple methods are

(i) Set the number of clusters to about $\sqrt{\frac{n}{2}}$, so that the expected number of points in each cluster is $\sqrt{2n}$

(ii) Use the Elbow Method

Evaluation of Clustering

- *Measuring clustering quality*
  - Assess how good the resulting clusters are

We have different methods based on weather ground truth are available or not.

Extrinsic Methods (when ground truth are available, also known as supervised methods)

Intrinsic Methods (when ground truth are not available, also known as unsupervised methods)

Extrinsic Methods

- Used when ground truth is available
- Find a quality score $Q(C, C_g)$
- In general a measure Q on clustering scheme quality is effective if it satisfies the following four essential criteria
  - Cluster homogeneity
  - Cluster completeness
  - Rag bag
  - Small cluster preservation

Cluster homogeneity

- This requires that the more pure the clusters in a clustering scheme are, the better the clustering scheme is i.e. a clustering quality measure, Q, respecting cluster homogeneity should give a higher score to C2 as compared to C1 i.e.
- $Q(C2, C_g) > Q(C1, C_g)$

  if C2 provides more homogenous clusters than C1

Cluster completeness

- Counterpart of cluster homogeneity
- If any two objects belong to the same category, according to the ground truth, then they should be assigned to the same cluster i.e.

$$Q\left(\square_2, \square_g\right) > Q\left(\square_1, \square_g\right)$$

if $\square_2$ provides more complete clusters than $\square_1$

Completeness measures whether **all points belonging to the same ground-truth class are assigned to the same cluster**.
We will use the **entropy-based completeness** formula:

$$\text{Completeness} = 1 - H(K|C)/H(K)$$

Where:

- $H(K|C)$ = conditional entropy of clusters given classes (weighted average)
- $H(K)$ = entropy of cluster distribution

Higher completeness → better clustering (1 = perfect completeness).

We have 6 points, with true class labels:

Class distribution: A=3, B=2, C=1
Total points = 6

Point	Class
p1	A
p2	A
p3	A
p4	B
p5	B
p6	C

Clustering Scheme 1 (Clustering 1)

Clusters obtained:

- Cluster X = {p1(A), p2(A)}
- Cluster Y = {p3(A), p4(B), p5(B)}
- Cluster Z = {p6(C)}

We will compute completeness for this clustering.

Step 1 – Compute entropy of clusters *H*(*K*) (marginal)

- Cluster X = {p1(A), p2(A)}
- Cluster Y = {p3(A), p4(B), p5(B)}
- Cluster Z = {p6(C)}

Cluster sizes:

|X|=2, |Y|=3, |Z|=1

Probabilities: P(X)=2/6=0.333, P(Y)=0.5, P(Z)=0.1667

$$H(K)=-(0.333\log_2 0.333+0.5\log 20.5+0.1667\log_2 0.1667)$$

Compute:

0.333 * (-1.585) = -0.528

0.5 * (-1) = -0.5

0.1667 * (-2.585) = -0.431

Sum negatives = -1.459, so

$$H(K)=1.459 \text{ bits}$$

Step 3 – Completeness for Clustering 1

Comp1=1−H(K|C)/H(K)=1−0.4591.459=1−0.3146=**0.6854**

Clustering Scheme 2 (Clustering 2)

Different clusters:

•Cluster X' = {p1(A), p2(A), p3(A)} (all A together)
•Cluster Y' = {p4(B), p5(B)} (all B together)
•Cluster Z' = {p6(C)}

Step 1 – H(K)*H*(*K*) for Clustering 2

Sizes: X'=3, Y'=2, Z'=1 → same probabilities as before (0.5, 0.333, 0.1667)
So H(K)=1.459 (identical, because cluster sizes same as Clustering 1)

Step 2 – H(K|C)$H(K|C)$ for Clustering 2

Class A (3 points): all in X' → distribution (1,0,0) → entropy = 0
Class B (2 points): all in Y' → entropy = 0
Class C (1 point): all in Z' → entropy = 0

Weighted H(K|C)=0 (perfect purity of classes to clusters)

Step 3 – Completeness for Clustering 2

Comp2=1−01.459=**1.0**

Which Clustering is Better?

- **Clustering 1** completeness = 0.685 → some points of same class (A) are split across clusters X and Y.
- **Clustering 2** completeness = 1.0 → every class is entirely contained in a single cluster.

Answer: Clustering 2 is better because it has higher (perfect) completeness.

Step 2 – Compute conditional entropy *H*(*K*|*C*)

Average over classes (weighted by class size):

For class A (3 points: p1,p2,p3):

- X contains 2 of A, Y contains 1 of A, Z contains 0 of A
- Distribution over clusters for class A: P(X|A)=2/3=0.667, P(Y|A)=1/3=0.333, P(Z|A)=0
- **Entropy for class A:**

H_A=−(0.667log20.667+0.333log20.333+0)

= - (0.667*(-0.585) + 0.333*(-1.585))

= - (-0.390 - 0.528) = -(-0.918) = **0.918**

For class B (2 points: p4,p5):

- X contains 0, Y contains 2, Z contains 0 → distribution (0, 1, 0) → entropy = 0 (certain)

For class C (1 point: p6):

- X=0, Y=0, Z=1 → certain → entropy = 0

Weighted H(K|C)*H*(*K*|*C*):

H(K|C)=36×0.918+26×0+16×0=0.5×0.918=0.459*H*(*K*|*C*)=63×0.918+62×0+61×0=0.5×0.918=0.459

Extrinsic Quality Measures (Ground Truth Known) – NEW EXAMPLE

Example dataset: 9 points

- True classes:

A, A, A, B, B, C, C, C, C

- Clustering result (k=3):
  - Cluster X = {p1(A), p2(A), p4(B)}
  - Cluster Y = {p3(A), p5(B), p6(C)}
  - Cluster Z = {p7(C), p8(C), p9(C)}

Step 1 – Homogeneity (each cluster contains only one class?)

- Cluster X: classes A(2), B(1) → impure
- Cluster Y: A(1), B(1), C(1) → impure
- Cluster Z: C(3) → pure

Step 2 – Quantify using conditional entropy

Total entropy of true classes:

$H(C) = -\ (3/9)\log_2(3/9) - (2/9)\log_2(2/9) - (4/9)\log_2(4/9)$

$= -0.333\cdot\log_2 0.333 - 0.222\cdot\log_2 0.222 - 0.444\cdot\log_2 0.444$

$= -0.333\cdot(-1.585) - 0.222\cdot(-2.17) - 0.444\cdot(-1.17)$

$= 0.528 + 0.482 + 0.519 = \mathbf{1.529}$

Conditional entropy H(C|K):

•Cluster X (size 3): class distribution A=2, B=1, C=0

$H(C|X) = -(2/3)\log(2/3) - (1/3)\log(1/3) = -0.667\cdot(-0.585) - 0.333\cdot(-1.585) = 0.390 + 0.528 = 0.918$

•Cluster Y (size 3): A=1,B=1,C=1 → uniform → $H = \log_2 3 = 1.585$

•Cluster Z (size 3): A=0,B=0,C=3 → H = 0

Weighted average:

$H(C|K) = (3/9)*0.918 + (3/9)*1.585 + (3/9)*0 = 0.306 + 0.528 + 0 = 0.834$

Homogeneity = 1 – H(C|K)/H(C) = 1 – 0.834/1.529 = 1 – 0.545 = 0.455

Step 3 – Completeness (all points of a class belong to same cluster?)

Symmetric calculation (swap roles of clusters and classes) gives **Completeness = 0.488**.

Interpretation: moderate quality.

Rag bag

- ‘Rag bag’ category containing objects that cannot be merged with other objects or clusters.
- Putting a heterogeneous object into a pure cluster should be penalized more than putting it into a rag bag.

$$Q\left(\square_{2},\square_{g}\right) > Q\left(\square_{1},\square_{g}\right)$$

if $\square_{2}$ places heterogeneous objects in one ‘rag bag’ cluster, where as $\square_{1}$ puts a heterogeneous object in some pure cluster.

Small cluster preservation

- Splitting a small category into pieces is more harmful than splitting a large category into pieces.
- A clustering quality measure Q should give a higher score to the clustering scheme preserving small clusters.

Intrinsic Methods

- Are applied when ground truth of the data set is not available.
- In general, these methods measure how well the clusters are separated.
- Silhouette coefficient of any object *o* is

$$s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}}.$$

where

$$a(o) = \frac{\sum_{o' \in C_i, o \neq o'} dist(o, o')}{|C_i| - 1}$$

$$b(o) = \min_{C_j : 1 \le j \le k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} dist(o, o')}{|C_j|} \right\}$$

The value of $s(o)$ is always between -1 and 1.

Intrinsic Measures – Silhouette Coefficient – NEW EXAMPLE

Dataset (2D):

Cluster A: (1,1), (2,2), (3,3)

Cluster B: (8,7), (9,8), (10,9)

Compute for point o = (2,2) in cluster A

•**a(o)** = avg distance to other points in same cluster A:

dist to (1,1) = $\sqrt{[(2-1)^2+(2-1)^2]} = \sqrt{2} = 1.414$

dist to (3,3) = $\sqrt{[(2-3)^2+(2-3)^2]} = \sqrt{2} = 1.414$

a(o) = (1.414+1.414)/2 = 1.414

•**b(o)** = min avg distance to points in other cluster (B):

Avg distance to B:

to (8,7) = $\sqrt{[(2-8)^2+(2-7)^2]} = \sqrt{(36+25)}=\sqrt{61}=7.81$

to (9,8) = $\sqrt{(49+36)}=\sqrt{85}=9.22$

to (10,9)= $\sqrt{(64+49)}=\sqrt{113}=10.63$

Avg = (7.81+9.22+10.63)/3 = 27.66/3 = 9.22

No other cluster, so b(o) = 9.22

•**s(o)** = (b – a) / max(a,b) = (9.22 – 1.414) / 9.22 = 7.806 / 9.22 = **0.846**

→ well-clustered.

Cluster average: compute for all points → A’s avg s ≈ 0.85, B’s avg s ≈ 0.84

→ global avg ≈ 0.845 → good clustering.

- When the silhouette coefficient value of an object *o* approaches 1, the cluster containing *o* is compact and *o* is far away from other clusters, which is the preferable case.
- To measure a cluster's fitness within a clustering, we can compute the average silhouette coefficient value of all objects in the cluster.
- To measure the quality of a clustering, we can use the average silhouette coefficient value of all objects in the data set.

Theoretical Part

- Definition of classification.
- Sensitivity, specificity, accuracy and error rate, precision and recall.
- Decision tree.
- Comparing Information gain, gain ratio and gini index
- Pre pruning and post pruning.
- Mutually exclusive and exhaustive rules
- Types of clustering techniques.
- Challenges in Divisive Hierarchical Clustering algorithms, and the suggested solutions.
- Properties (strengths/weaknesses) of K-means, K-medoids, CLARA, CLARANS and BIRCH.
- Defining distance between clusters.
- Difference between agglomerative and divisive clustering.
- Properties of clustering features.